

Module3Lab2

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1 Module 3 Hands-On Problem 2.1: Outliers in the Crime Data

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Course: CS 82A

Repo: <https://github.com/raulmckay-blip/cs82a-portfolio>

1.1 Overview

This lab looks at a real crime dataset (`usarrests.csv`) to find values that can't actually be true. It has one row for each US state, with arrest rates per 100,000 people for murder and assault, and the percent of each state's population living in urban areas.

There are two kinds of bad values to check for:

- **Impossible:** the value breaks the definition of the measurement. A percent over 100 is a good example, since a percent can never go higher than 100.
- **Extreme:** the value is just unusual, not against any rule. It's a value that could really happen, just far from the rest.

If a value is impossible, it's an error, so I fix it. If a value is extreme, it's real data, so I keep it.

```
[13]: import pandas as pd
import numpy as np

df = pd.read_csv("usarrests.csv")
df = df.rename(columns={"Unnamed: 0": "State"})
print(df.shape)
df.head()
```

(50, 4)

```
[13]:
```

	State	Murder	Assault	UrbanPop
0	Alabama	13.2	236.0	58
1	Alaska	10.0	263.0	48
2	Arizona	8.1	294.0	80
3	Arkansas	8.8	190.0	50
4	California	9.0	276.0	91

1.2 Summary Statistics

```
[14]: df.describe()
```

```
[14]:
```

	Murder	Assault	UrbanPop
count	50.00000	49.00000	50.00000
mean	7.78800	182.183673	74.20000
std	4.35551	130.877435	73.40828
min	0.80000	45.00000	6.00000
25%	4.07500	109.00000	53.25000
50%	7.25000	159.00000	66.00000
75%	11.25000	249.00000	77.75000
max	17.40000	879.00000	570.00000

1.3 Impossible Values

```
[15]: df[df["UrbanPop"] > 100]
```

```
[15]:
```

	State	Murder	Assault	UrbanPop
14	Iowa	2.2	56.0	570

```
[16]: num_cols = df.select_dtypes("number").columns
df[(df[num_cols] < 0).any(axis=1)]
```

```
[16]: Empty DataFrame
Columns: [State, Murder, Assault, UrbanPop]
Index: []
```

1.4 Extreme Values (IQR Check)

```
[17]: for col in num_cols:
    q1, q3 = df[col].quantile([0.25, 0.75])
    upper = q3 + 1.5 * (q3 - q1)
    lower = q1 - 1.5 * (q3 - q1)
    flagged = df[(df[col] > upper) | (df[col] < lower)]
    if len(flagged) > 0:
        print(f"{col}: outside {lower:.1f} to {upper:.1f}")
        display(flagged)
```

Assault: outside -101.0 to 459.0

```
State Murder Assault UrbanPop
39 South Carolina 14.4 879.0 48
```

UrbanPop: outside 16.5 to 114.5

```
State Murder Assault UrbanPop
14 Iowa 2.2 56.0 570
31 New York 11.1 254.0 6
```

1.5 Handling the Impossible Value

```
[18]: df.loc[df["UrbanPop"] > 100, "UrbanPop"] = np.nan
      df.describe()
```

```
[18]:
```

	Murder	Assault	UrbanPop
count	50.00000	49.00000	49.00000
mean	7.78800	182.18367	64.08163
std	4.35551	130.87743	16.59296
min	0.80000	45.00000	6.00000
25%	4.07500	109.00000	53.00000
50%	7.25000	159.00000	66.00000
75%	11.25000	249.00000	77.00000
max	17.40000	879.00000	91.00000

1.6 Flagged Values

Looking at `df.describe()` and the IQR check, I found four values worth flagging: one impossible, two extreme, and one missing value that was already in the data before I touched it.

1. Iowa, UrbanPop = 570. Impossible. UrbanPop is a percentage, so it can't go above 100. A value of 570 is more than five times too high, which is also why the column's max (570) is so much bigger than its 75th percentile (77.75) in the summary table. This is most likely a data entry mistake, maybe an extra digit or a decimal point in the wrong spot. I replaced it with NaN instead of deleting the whole row so Iowa's other numbers are kept.

2. South Carolina, Assault = 879. Extreme. The IQR check flags anything above 459 as outside the normal range for this column, and 879 goes way past that. It's almost double the next highest state, Florida, at 335. Even so, it's still a real, possible arrest rate. Nothing about the measurement rules out a number that high. I kept it because taking out real extreme values would hide actual differences between states.

3. New York, UrbanPop = 6. Extreme. The same IQR check flags anything below 16.5 as unusual, and 6 is well under that. It's still a real percentage, somewhere between 0 and 100, so it doesn't count as impossible. But a 6 percent urban population for a state like New York doesn't match what I'd expect in real life. I kept the value, but I'm noting it here as one I'd want to double check against a real source before using it in any analysis.

4. Georgia, Assault. Missing, not impossible or extreme. This one doesn't really fit the other two categories. It's not a bad value, it's just not there. The count for **Assault** in the summary table is 49 instead of 50, and checking that showed Georgia's **Assault** cell is blank in the raw file. I didn't cause this by cleaning the data. It was already missing before I changed anything, so I left it as NaN instead of guessing what it should be.